Centre Number	Index Number	Name	Class
S3016		Suggested Solutions	

RAFFLES INSTITUTION
2026 Y6 Science Practical Practice Session

PHYSICS **9478/04**
Higher 2

40 minutes

You must answer on the question paper.

INSTRUCTIONS
- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your index number, name and class in the boxes at the top of the page.
- Shade the lozenges in the Student ID box below according to your Student ID.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.
- You may use a spreadsheet to process and analyse data.
- Write the details of the shift and laboratory in the boxes provided.
- You will be allowed to work with the apparatus for a maximum of 1 hour 15 minutes for each section.

INFORMATION
- The number of marks for each question or part question is shown in brackets [ ].

Shift
Laboratory

Student ID

2 6

For Examiner's Use	
1	/ 14
Deduction	
Total	/ 14

This document has **5** pages.

In this experiment, you will investigate the oscillation of a load attached to a spring.

Set up the experiment as shown in Figure 1.1, using a total mass of 200 g for the load.

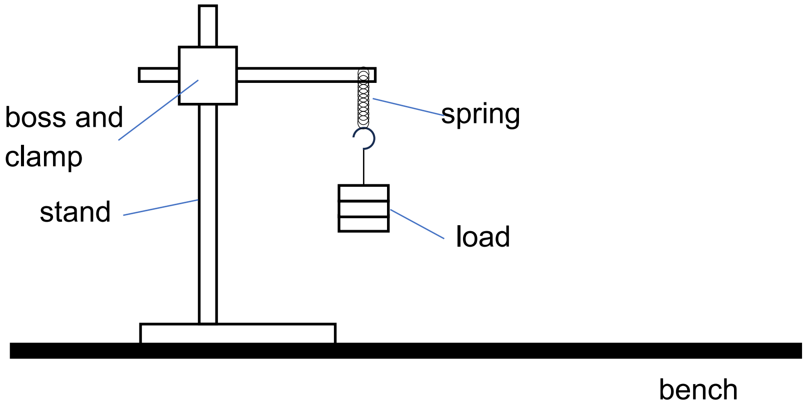


Fig. 1.1

(a) **(i)** Pull the load down and release it such that the load oscillates vertically.

Determine the period T of the oscillation.

$$T = \frac{23.34 + 23.24}{2 \times 40}$$
$$= 0.5823 \text{ s}$$

T = 0.5823 s .... [1]

(ii) Calculate the angular frequency ω of the oscillation, where $\omega = \frac{2\pi}{T}$.

$$\omega = \frac{2\pi}{0.5823}$$
$$= 10.79 \text{ rad s}^{-1}$$

ω = 10.79 rad s^{-1} .... [1]

(b) The oscillation in **(a)** is observed to be damped. A student attempted to provide a periodic force to drive the oscillation, such that the load exhibits simple harmonic motion.

The student repeats **(a)(i)** and **(a)(ii)**, this time using a digital device to record the time and velocity of the load during the oscillation.

The data is recorded in the digital file: datasetQ1b.xlsx.

(i) Use the student's data to estimate the amplitude A of the oscillation.

Using the average of two quarter-cycle areas,

$$A = \frac{0.04985}{2}$$
$$= 0.0249 \text{ m}$$

A = 0.0249 m [1]

(ii) Explain how you obtained the value of A, and state how the uncertainty in the value of A obtained can be reduced.

The amplitude of the oscillation can be found from the area under a quarter cycle of the velocity-time graph.

The uncertainty in the value of A can be reduced by taking the average of two or more quarter cycles. [2]

OR: The amplitude of the oscillation can be calculated using $A = \frac{v_0}{\omega}$, where v_0 is maximum magnitude of the velocity of the load which can be found from the data set and $\omega = \frac{2\pi}{T}$.

The uncertainty in the value of A can be reduced by using the average of two or more values of v_0 and T.

(iii) For an object undergoing simple harmonic motion, the acceleration a is related to the position x by the equation $a = -\omega^2 x$.

Explain how the acceleration a and the position x can be determined from the student's data.

The acceleration *a* can be determined by using the SLOPE function in Excel to obtain the gradient at a point on the velocity-time graph.

The position *x* can be determined by the area under the velocity-time graph using the trapezium rule.

[2]

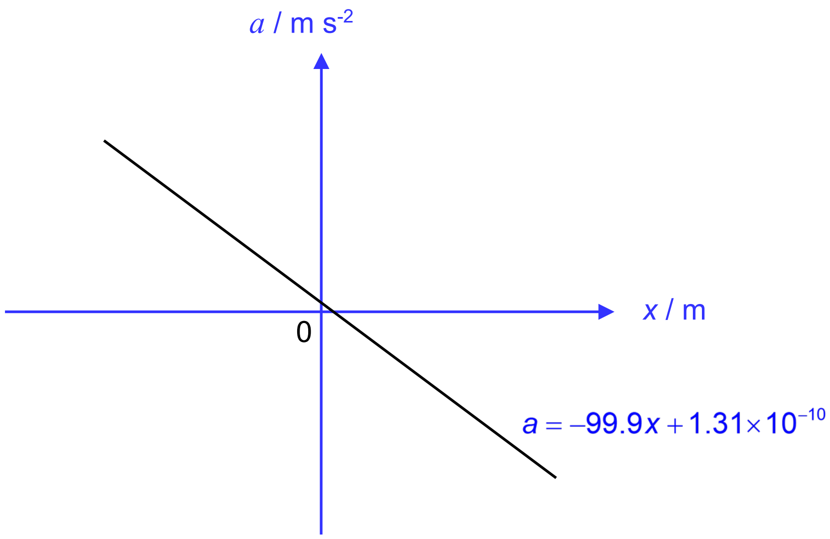
(iv) Plot a suitable graph using a spreadsheet to determine the angular frequency ω of the oscillation.

Sketch the graph in the space below. Write down the equation of the trendline.

Plot a graph of a against x where $\text{gradient} = -\omega^2$ (and vertical intercept should be 0)

OR

Plot a graph of x against a where $\text{gradient} = -\dfrac{1}{\omega^2}$ (and vertical intercept should be 0)



$\omega^2 = 99.9$

$\omega = 9.99$ rad s^{-1}

[3]

(c) It is suggested that the angular frequency ω, the spring constant k and the load m are related by the relationship

$$\omega = \sqrt{\frac{k}{m}}$$

Plan an investigation to show that the relationship is valid.

You are not required to do this experiment.

Your account should include:

- your experimental procedure
- any precautions that should be taken to improve the accuracy and/or safety of the investigation
- how you would use your results to show that the relationship is valid.

1. Set up the apparatus as shown in Fig. 1.1. The same spring is used for all different sets of readings for different masses *m*.
2. Using a total mass *m* of 150 g for the load, pull the load down and release it such that it oscillates vertically.
3. Take the timing with a stopwatch for a suitable number of oscillations *n* such that the total time *t* taken is at least 20 s. Repeat the experiment and determine the average time $\langle t \rangle$.
4. Determine the period *T* of the oscillation using $T = \frac{\langle t \rangle}{n}$.
5. Repeat steps 2 – 4 for different masses for a total of 8 sets of readings for *m* and *T* using the same spring.
6. Use $\omega = \frac{2\pi}{T}$ to determine ω^2 for the different masses *m*. Calculate the corresponding values for $\frac{1}{m}$.
7. Plot a graph of ω^2 against $\frac{1}{m}$. If a straight-line graph with zero vertical intercept (or close to zero vertical intercept) is obtained, then the relationship is valid.
8. Pull the load is vertically downwards with a small displacement to ensure correct mode of oscillations. Restart the experiment if sideways motion is observed.

   OR:
9. Load the spring starting with the smallest load *m* and increase the load gradually. If the spring does not return to its original length after the load is removed, the spring is overloaded.

[4]

End of Practice